

Manjula Herath

BDS, MD (Oral & Maxillofacial Surgery)

Consultant OMF Surgeon & Doktorand (PhD), Malmö University

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Professional Profile

01

A Consultant Oral & Maxillofacial Surgeon and doctoral researcher practising a **digital-first approach** — every complex case planned in Virtual Surgical Planning (VSP) and executed through CAD/CAM workflows. Subspecialty focus in **head-and-neck oncology**, **cleft lip & palate** and **microvascular free-flap reconstruction**, within the full spectrum of OMF surgery — orthognathic, salivary-gland and parotid, TMJ, and the complete implant range to zygomatic implants and full-mouth rehabilitation. Co-developer of an internationally adopted **open-source surgical-planning platform** and developer of in-house patient-specific titanium and PEEK implant workflows. Certified postgraduate Trainer and Examiner in OMFS at the Post Graduate Institute of Medicine (PGIM), Colombo, and Key Opinion Leader for Dentium in Sri Lanka — with a sustained record of teaching, supervision and peer-reviewed publication, including service as Local Faculty for an AO CMF surgical course.

Career Snapshot

02

6,343 Patient episodes managed in calendar year 2025 as Sole Consultant	14 Microvascular free-flap reconstructions in 2025 (radial forearm, fibula, DIEP)	175 Fully-digital (VSP) surgical cases delivered to date across the subspecialties
4 Countries with documented surgical cases using his open-source planning platform	16 Publications & presentations including a co-authored national textbook	President Government Dental Surgeons' Association, Sri Lanka (2024–2025)

Current Appointment & Clinical Activity

03

Consultant Oral & Maxillofacial Surgeon

Present

District General Hospital, Nuwara Eliya · Ministry of Health, Sri Lanka · Sole Consultant

- Sole Consultant delivering the district's complete maxillofacial service — trauma, head-and-neck oncology and reconstruction, orthognathic and TMJ surgery.
- Established advanced microvascular and fully-digital reconstruction in a peripheral government hospital, delivering tertiary-level complexity outside a tertiary centre.

HEAD & NECK ONCOLOGY

239 new cancers

127 managed surgically · 147 oncology referrals coordinated; ablative resection with primary reconstruction.

MICROVASCULAR RECONSTRUCTION

14 free flaps

12 radial-forearm · 1 free fibula · 1 DIEP — the majority VSP-planned with 3D-printed cutting guides.

CLEFT LIP & PALATE

Multidisciplinary regional service

Primary & secondary cleft surgery and alveolar-cleft reconstruction (doctoral focus); member of the cleft & craniofacial team.

Supporting activity from the **Monthly Return of the OMFS Units**, Ministry of Health, Sri Lanka (DGH Nuwara Eliya; 01 Jan – 31 Dec 2025).

SERVICE VOLUME		MAXILLOFACIAL TRAUMA	
Total patient episodes	6,343	Facial-bone fractures	224
Inpatients managed	596	Dento-alveolar trauma	834
Medico-legal cases	321	Soft-tissue injuries	748
ORAL POTENTIALLY-MALIGNANT DISORDERS		RECONSTRUCTIVE & GLANDULAR	
Oral submucous fibrosis	244	Orthognathic surgeries	9
Leukoplakia / erythroplakia	165 / 106	Salivary-gland procedures	188
Erosive lichen planus	66	Cyst enucleations / TMJ disorders	110 / 263

Routine workload (abridged): dento-alveolar and minor oral surgery — extractions, surgical tooth removal and other minor procedures — exceeded **2,150** cases across the year alongside the specialist caseload above.

Digital Surgery, VSP & CAD/CAM

04

A “Scan – Plan – Execute” digital surgical practice

Every complex reconstruction is planned pre-operatively as a digital simulation and executed with **3D-printed patient-specific instruments**, custom selective-laser-melted titanium and PEEK implants, and augmented/virtual-reality guidance. **175 fully-digital surgical cases** completed to date — 82 head-and-neck cancer reconstructions, 28 dental-implant, 22 orthognathic, 21 trauma, 11 craniofacial, 9 TMJ and 2 thyroid-eye-disease corrections — a sustained, fully-digital reconstructive caseload spanning every subspecialty.

SlicerBoneReconstructionPlanner CO-DEVELOPER · OPEN-SOURCE

Free, open-source 3D Slicer extension for VSP of mandibular reconstruction with fibula free flap, with automated design of resection and cutting guides (with M. Dominguez and Prof. A. Lasso, PerkLab, Queen's University). Cuts per-case planning and guide cost from **~USD 8,000** commercially to **~LKR 1,500** in materials; documented surgical cases across **four countries** — Sri Lanka, Malaysia, Argentina and Hungary.

Patient-Specific Implant & Guide Workflow DESIGN & MANUFACTURE

End-to-end CAD/CAM pipeline designing and producing patient-specific titanium (SLM) and PEEK implants and high-precision resin surgical guides for cranio-maxillofacial reconstruction, manufactured in-house under strict quality control — a professional print-failure rate held below **0.5% per resin bottle**.

3D-Printed Ventilators CONTRIBUTOR

Open-source contribution to 3D-printed ventilator development at Malmö University during the COVID-19 response — applying additive-manufacturing expertise to critical-care equipment shortages.

Computational toolchain: ProPlan CMF · Materialise Mimics · Mimics Enlight CMF · NemoStudio · 3D Slicer · Blender · FreeCAD · selective laser melting · resin & PEEK 3D printing.

Med-Tech Translation & Manufacturing

Co-Founder, Protomed CMF & Clear Plus — translating academic digital-design pipelines into commercial clinical manufacturing, producing patient-specific cranio-maxillofacial implants and orthodontic appliances under rigorous quality-control standards.

Education & Qualifications

05

Doktorand (Doctoral Researcher / PhD candidate) — Oral & Maxillofacial Surgery

2020 — Present

Malmö University, Sweden · Faculty of Odontology

- Doctoral research on alveolar-cleft reconstruction, autogenous bone harvesting and guided bone regeneration with PEEK, alongside open-source digital-surgery tool development.

Doctor of Medicine (MD) in Oral & Maxillofacial Surgery

2014 — 2019

University of Colombo · Post Graduate Institute of Medicine (PGIM)

Bachelor of Dental Surgery (BDS)

2003 — 2009

University of Peradeniya · Distinctions in Oral Pathology and Restorative Dentistry

Professional Experience

06

Consultant Oral & Maxillofacial Surgeon

Present

District General Hospital, Nuwara Eliya · Ministry of Health, Sri Lanka · Sole Consultant

Specialist Trainee (Board-Certification Training), OMFS

Jun 2019 — Feb 2021

Malmö University, Sweden · Overseas specialist training

Specialist Trainee (Registrar / Senior Registrar), OMFS

Jan 2015 — May 2019

Post Graduate Institute of Medicine, University of Colombo · MD (OMFS) training programme

Teaching, Training & Faculty Appointments

07

EXAMINER & POSTGRADUATE TRAINER

Certified Examiner & Postgraduate Trainer in OMFS

Ongoing

Post Graduate Institute of Medicine (PGIM), University of Colombo

- Trained and credentialed as a postgraduate Trainer and as an Examiner for the MD (OMFS) programme.
- As Secretary of the Speciality Board in OMFS (2022–2024), coordinated training, conducted examinations and led development of the MD (OMFS) curriculum; currently a Member of the Board of Study in OMFS.

SURGICAL-COURSE FACULTY & INTERNATIONAL PANELLIST

Local Faculty — AO CMF Course: Surgical Techniques on Condyle Fractures with Anatomical Specimens

18–19 Dec 2021

AO CMF (AO Foundation) · Faculty of Dental Sciences, University of Peradeniya, Sri Lanka

- Led cadaveric practical stations — surgical approaches to the TMJ, fixation of sub-condylar fractures, surgical access to the condylar head and the high-condylar shave.

International Faculty / Panellist — IndoCleftCon: Challenges & Complications in Cleft Palate Surgery

2022

IndoCleftCon · Kochi, India

Faculty — Oral & Digital Dentistry

2021

Oral & Digital Dentistry · Colombo, Sri Lanka

RECENT HANDS-ON MASTER CLASSES & WORKSHOPS CONVENED

A representative snapshot — convenes hands-on surgical, CBCT and digital-workflow master classes on an ongoing basis.

"Master Your CBCT — 3D Slicer Workshop"

12 Jun 2026

3rd Dentistree Congress 2026 · Cinnamon Grand, Colombo · Government Dental Surgeons' Association

- CBCT anatomy & DICOM workflows, inferior-alveolar-nerve identification for impacted third molars, implant bone-assessment and safety zones, and STL / surgical-guide generation with 3D printing.

"Foundations of Dental Implantology — From Incision to Restoration" (Hands-on skills lab)

13 Jun 2026

3rd Dentistree Congress 2026 · Cinnamon Grand, Colombo · with Dr Chanaka Dissanayake

"The Art of Removing Impacted Lower Third Molars with Confidence"

2025

2nd Dentistree Congress 2025 & Dentex Trade Fair · with Dr Nadun Dhammika

- 3D surgical anatomy, CBCT segmentation of the inferior alveolar nerve, step-by-step technique, intra-operative complication management and post-operative care.

INVITED LECTURES

"CAD/CAM Workflow and Manufacturing in Maxillofacial Reconstruction"

6–7 Apr 2024

Annual Academic Sessions — College of Dentistry and Stomatology of Sri Lanka · Water's Edge Hotel, Colombo

POSTGRADUATE ACADEMIC SUPERVISION — MASTER'S THESES, MALMÖ UNIVERSITY

Samah Khalil & Farah Shkair

2021

Ex-vivo accuracy of visualising peri-implant bone thickness using CBCT and intraoral scanning

Pegah Mirdamadian & Raha Salahshour Nargi

2021

Grafting materials for alveolar cleft reconstruction — a systematic review

Alise Austruma & Charlotte Nicklasson

2022

Comparison of three autogenous bone-harvesting methods on bone volume

CLINICAL TEACHING & MENTORING

- Provided supervision, training and support to senior registrars, registrars, senior house officers and house officers; taught dental surgeons, nurses and paramedical students across the wider surgical team.
- Member of the multidisciplinary cleft lip & palate clinic, contributing to clinical decision-making and the management of patients with craniofacial deformities.

Research Activities

08

Randomised controlled trial — parallel-group comparison of two donor sites (mandibular ramus vs anterior iliac crest) for alveolar cleft reconstruction; guided bone regeneration with PEEK.

Systematic review & meta-analysis — grafting materials for alveolar cleft reconstruction (PROSPERO CRD42020197451).

Open-source software modules — fibula free-flap reconstruction planning (SlicerBoneReconstructionPlanner / SlicerIGT).

Doctoral thesis (ongoing) — Malmö University: digital workflows and reconstruction in oral & maxillofacial surgery.

Selected Publications

09

TEXTBOOK

B Herath M, et al. Handbook for Dental Interns. 1st ed. Ministry of Health, Sri Lanka; **2019**. ISBN 978-955-3666-46-8.

PEER-REVIEWED JOURNAL ARTICLES

- 1 Toia M, Stocchero M, Jinno Y, et al. Inter-implant distance correlated to different preparation protocol on cortical bone: an animal study. *BMC Oral Health*. **2025**;25:1512.

- 2 Maisi S, Dominguez M, Gilong PC, Kiong CT, Hajam S, Badruddin AFA, **Herath M**, et al. In-house virtual surgical planning for mandibular reconstruction with fibula free flap: case series and literature review. *Annals of 3D Printed Medicine*. **2023**;10:100109.
- 3 Stocchero M, Jinno Y, Toia M, Ahmad M, Galli S, Papia E, **Herath M**, et al. Effect of drilling preparation on immediately loaded implants: an in vivo study in sheep. *Int J Oral Maxillofac Implants*. **2023**;38(3).
- 4 Ratnayake A, Nanayakkara S, **Herath M**. A case of displaced mandibular reconstruction plate and the airway. *Anaesthesia, Pain & Intensive Care*. **2023**;27(1):139–141.
- 5 Samaraweera W, **Herath M**, Pathirage SL. Management of impacted maxillary canines. *Sri Lanka Dental Journal*. **2020**;50(03):93–102.
- 6 **Herath MPB**, Siriwardena BSMS, Jayasinghe RD, Dias DK. Oropharyngeal epignathus with partial facial duplication: report of a rare case. *Int J Oral Maxillofac Surg*. **2017**;47(4):442–444.
- 7 Jayasinghe R, Ekanayake R, **Herath MPB**, Sitheequ MAM, Amarasinghe H, Tilakaratne WM. Intralesional corticosteroids as a treatment in restricted mouth opening in oral submucous fibrosis. *Oral Surg Oral Med Oral Pathol Oral Radiol (OOOO)*. **2016**;122(2):224–231.

INVITED PROFESSIONAL ARTICLE

- ★ **Herath M**. Digital transformation in surgery: an oral and maxillofacial surgeon's experience and universal applications. *The SLMA Monthly (Sri Lanka Medical Association)*. **2025**;18(11):6–9.

CONFERENCE PRESENTATIONS & PROCEEDINGS

- 1 **Herath M**, Delpechitra D, Dias DK. Elective versus therapeutic neck dissection in node-negative squamous cell carcinoma of the tongue and buccal mucosa. *3rd Asian Maxillofacial Trainees Conference*; **2017**.
- 2 **Herath M**, Dissanayaka A, Perera B, Dias DK. Modified unilateral buccinator myomucosal flap technique for correction of velopharyngeal insufficiency. *5th Biennial Academic Sessions, PGIM*; **2016**. DOI:10.4038/jpgim.8128.
- 3 **Herath M**, Siriwardena BSMS, Dias DK. Epignathus — first description in Sri Lankan medical literature. *13th Annual Scientific Sessions, SLAOMS*; **2016**.
- 4 **Herath M**, Dias DK, Pathirage SL, Samaraweera W. Surgically assisted rapid palatal expansion (SARPE) in a case of constricted maxilla. *Int. Orthodontic Conference, Sri Lanka Orthodontic Society*; **2016**. DOI:10.13140/RG.2.1.2584.2808.
- 5 **Herath M**, Dias DK, Pathirage SL, Samaraweera W. Surgical exposure and orthodontic traction of a labially impacted maxillary canine. *Int. Orthodontic Conference, Sri Lanka Orthodontic Society*; **2016**. DOI:10.13140/RG.2.1.1011.4163.
- 6 Siriwardena BSMS, Jayathilake DSY, **Herath MPB**, Jayasooriya PR, Tilakaratne WM. Demographic and histological variations of oral squamous cell carcinoma in Sri Lanka. *32nd Asia Pacific Dental Congress*; **2010**.
- 7 **Herath MPB**, Jayasooriya PR, Liyanage RLPR. Clinico-pathological comparison of fibrous epulis and ossifying fibrous epulis. *Annual Scientific Session, Sri Lanka Dental Association (Kandy Branch)*; **2010**.

Continuous Professional Development

10

AO CMF AO FOUNDATION CRANIOMAXILLOFACIAL COURSES

Principles in Craniomaxillofacial Fracture Management
— AO CMF Course, Colombo (2014)

Advances in Mandible & Midface Reconstruction — AO
CMF Course, Colombo (2015)

Advances in Surgical Correction of Facial Skeletal Deformities — AO CMF Symposium, Colombo (2016)

Management of Complex Facial Trauma (Upper & Middle Third) — AO CMF Seminar, Colombo

DOCTORAL COURSEWORK — MALMÖ UNIVERSITY

Research Ethics & Responsible Conduct in Research (2021)

Health Innovation & Entrepreneurship — School of Health Innovation (Oslo · Karolinska · Copenhagen · NTNU; Novo Nordisk Foundation)

Academic Writing & Publishing (2021)

OMFS SKILLS WORKSHOPS & COURSES

Microvascular Surgery Workshop Series (2015)

Reconstructive Surgery, Module 02 — IAOMS Regional Workshop, Galle (2016)

Orthognathic & Reconstructive Surgery (2015, 2017)

Facial Aesthetics & FACE Course “Around the Orbit” (2015)

Management of Salivary Gland Diseases (2015)

Professional Leadership & Memberships

President — Government Dental Surgeons' Association, Sri Lanka (2024–2025)

General Secretary — Government Dental Surgeons' Association (2022–2024); Executive Committee Member (current)

Joint Secretary & former Editor — College of Oral & Maxillofacial Surgeons of Sri Lanka (Editor 2021–2024)

Secretary & Board Member — Speciality Board / Board of Study in OMFS, PGIM (Secretary 2022–2024)

Council Member — Sri Lanka Dental Association

Awards & Honours

First Runner-up, Oral Presentations — 5th Biennial Academic Sessions, PGIM (2017)

Distinction in Oral Pathology — University of Peradeniya (2009)

Referees

Prof. Emeritus Lakshman Samaranayake

Editor-in-Chief, International Dental Journal; former Dean of Dentistry, University of Hong Kong & University of Queensland

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Supporting the Learning of Others (2021)

Communicating Research (2021)

Theory of Science (2021)

Secondary Management of Cleft Lip & Palate (2015, 2016)

Craniofacial Developmental Anomalies & Challenges (2016)

National Trauma Management Course (IATSIC) — Colombo

Professionalism & Ethics in Medical Practice — PGIM, Colombo

Editor — Sri Lanka Association of Cleft Lip, Palate & Craniofacial Anomalies

Peer Reviewer — Scientific Reports (Springer Nature); International Dental Journal (Elsevier)

Key Opinion Leader — Dentium (dental-implant systems), Sri Lanka

Member — AO CMF (AO Foundation); Sri Lanka Association of Oral & Maxillofacial Surgeons

Distinction in Restorative Dentistry — University of Peradeniya (2009)

Dr D. K. Dias

Consultant Oral & Maxillofacial Surgeon; former Head, Department of OMF Surgery & Regional Cleft Centre, Karapitiya Teaching Hospital, Galle

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